12.3. Metric-by-Metric Performance Comparison: PE vs. WPE on raw and curated datasets

We then added the performances on curated datasets to the previous comparison between PE and WPE on raw datasets. We can see that the performances are really improved with curated datasets, especially for the Hall dataset.

12.3.1. Balanced Accuracy Comparison

Target Group	Evaluation Context / Centroids	PE Acc.	WPE raw Acc.	WPE curated Acc.	Acc. Δ
Within-Dataset Clustering (Self-Trained Centroids)					
DBT2/NDBT2	K-Means (Self)	0.98	0.97	**1.00**	+0.03
CGMacros + Dexcom	K-Means (Self)	0.79	0.86	**0.96**	**+0.10**
CGMacros + Libre	K-Means (Self)	0.90	0.86	**1.00**	**+0.14**
Average CGMacros	K-Means (Self)	0.845	0.86	**0.98**	**+0.12**
DBT2/NDBT2 + CGMacros	K-Means (Self)	0.93	0.91	**0.99**	+0.08
Hall	K-Means (Self)	0.85	0.79	**0.88**	+0.09
Cross-Dataset Centroid Validation Transfer					
CGMacros + Dexcom	DBT2/NDBT2	0.83	0.86	**0.96**	**+0.10**
CGMacros + Libre	DBT2/NDBT2	0.93	0.90	**1.00**	**+0.10**
Hall	DBT2/NDBT2 + CGMacros	0.86	0.75	**0.88**	**+0.13**

Table 28: Balanced Accuracy comparison between PE, WPE raw, and WPE curated feature spaces.

12.3.2. F1-Score Comparison

Target Group	Evaluation Context / Centroids	PE F1	WPE raw F1	WPE curated F1	F1 Δ
Within-Dataset Clustering (Self-Trained Centroids)					
DBT2/NDBT2	K-Means (Self)	0.98	0.98	**1.00**	+0.02
CGMacros + Dexcom	K-Means (Self)	0.81	0.86	**0.96**	**+0.10**
CGMacros + Libre	K-Means (Self)	0.90	0.86	**1.00**	**+0.14**
Average CGMacros	K-Means (Self)	0.855	0.86	**0.98**	**+0.12**
DBT2/NDBT2 + CGMacros	K-Means (Self)	0.93	0.91	**0.99**	+0.08
Hall	K-Means (Self)	0.93	0.87	**0.98**	**+0.11**
Cross-Dataset Centroid Validation Transfer					
CGMacros + Dexcom	DBT2/NDBT2	0.81	0.86	**0.96**	**+0.10**
CGMacros + Libre	DBT2/NDBT2	0.93	0.89	**1.00**	**+0.11**
Hall	DBT2/NDBT2 + CGMacros	0.95	0.92	**0.98**	+0.06

Table 29: F1-Score performance comparison between PE, WPE raw, and WPE curated.

12.4. Sensitivity Comparison

Target Group	Evaluation Context / Centroids	PE Sens.	WPE raw Sens.	WPE curated Sens.	Sens. Δ
Within-Dataset Clustering (Self-Trained Centroids)					
DBT2/NDBT2	K-Means (Self)	0.95	1.00	**1.00**	0.00
CGMacros + Dexcom	K-Means (Self)	0.87	0.80	**0.92**	**+0.12**
CGMacros + Libre	K-Means (Self)	0.87	0.80	**1.00**	**+0.20**
Average CGMacros	K-Means (Self)	0.87	0.80	**0.96**	**+0.16**
DBT2/NDBT2 + CGMacros	K-Means (Self)	0.92	0.88	**0.97**	+0.09
Hall	K-Means (Self)	0.89	0.79	**1.00**	**+0.21**
Cross-Dataset Centroid Validation Transfer					
CGMacros + Dexcom	DBT2/NDBT2	0.73	0.80	**0.92**	**+0.12**
CGMacros + Libre	DBT2/NDBT2	0.87	0.80	**1.00**	**+0.20**
Hall	DBT2/NDBT2 + CGMacros	0.92	0.89	**1.00**	**+0.11**

Table 30: Sensitivity (True Positive Rate) comparison between PE, WPE raw, and WPE curated.

12.5. Specificity Comparison

Target Group	Evaluation Context / Centroids	PE Spec.	WPE raw Spec.	WPE curated Spec.	Spec. Δ
Within-Dataset Clustering (Self-Trained Centroids)					
DBT2/NDBT2	K-Means (Self)	0.91	0.95	**1.00**	+0.05
CGMacros + Dexcom	K-Means (Self)	0.71	0.93	**1.00**	+0.07
CGMacros + Libre	K-Means (Self)	0.93	0.93	**1.00**	+0.07
Average CGMacros	K-Means (Self)	0.82	0.93	**1.00**	+0.07
DBT2/NDBT2 + CGMacros	K-Means (Self)	0.94	0.94	**1.00**	+0.06
Hall	K-Means (Self)	0.80	**0.80**	0.75	−0.05
Cross-Dataset Centroid Validation Transfer					
CGMacros + Dexcom	DBT2/NDBT2	0.93	0.93	**1.00**	+0.07
CGMacros + Libre	DBT2/NDBT2	1.00	1.00	**1.00**	0.00
Hall	DBT2/NDBT2 + CGMacros	0.80	0.60	**0.75**	**+0.15**

Table 31: Specificity (True Negative Rate) comparison between PE, WPE raw, and WPE curated.